

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 15

Build an Enterprise-Style AWS Organizations OU and Account Structure

AWS Organizations | OUs | Multi-Account Governance

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 13, 2026
STATUS	Complete

Executive Summary

Began building an enterprise-style AWS Organizations structure under the organization root. Created top-level Organizational Units for **Infrastructure**, **Workloads**, **Exceptions**, **Sandbox**, **Onboarding**, **Nursery**, **Suspended**, and **IncidentResponse**. Also created two new security-focused AWS accounts, **IAM** and **SecurityOperations**, and placed them with the existing SecurityAudit account under the **Security** OU.

Business Problem

A single flat AWS account structure becomes difficult to secure and operate as more services, teams, and environments are added. Enterprise AWS environments use accounts and Organizational Units as isolation and governance boundaries. This structure reduces blast radius, supports Service Control Policy targeting, separates operational responsibilities, and creates account lifecycle locations for onboarding, exceptions, incident response, sandbox activity, and account suspension.

Objectives

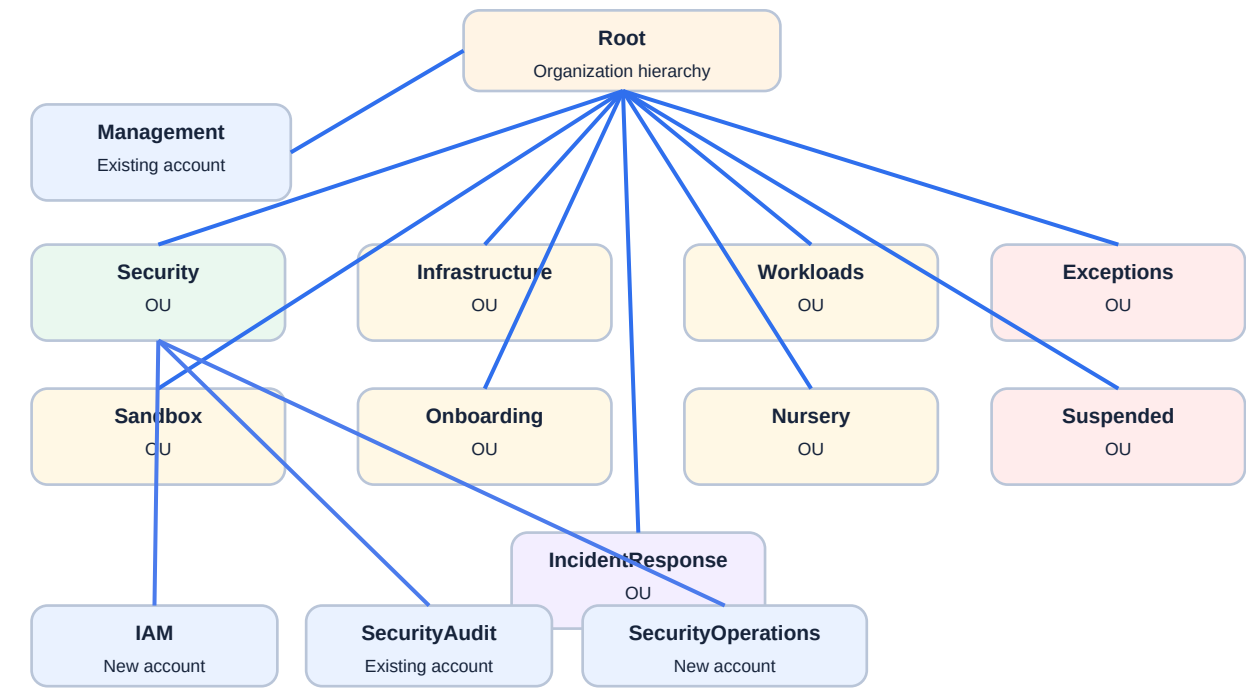
- Expand the organization root into a more enterprise-style hierarchy.
- Create dedicated OUs for infrastructure, workloads, exceptions, sandbox activity, onboarding, account nursery, suspended accounts, and incident response.
- Create an IAM account for future centralized identity administration.
- Create a SecurityOperations account for future security tooling that can make changes.
- Keep SecurityAudit in the Security OU as the audit and assessment account.
- Document the purpose and governance value of each OU and security account.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	AWS Organizations
Existing OU	Security
New Top-Level OUs	Infrastructure, Workloads, Exceptions, Sandbox, Onboarding, Nursery, Suspended, IncidentResponse
Existing Security Account	SecurityAudit
New Security Accounts	IAM and SecurityOperations
Account Placement	IAM, SecurityAudit, and SecurityOperations placed under Security OU
Purpose	Enterprise-style multi-account hierarchy and governance foundation

Evidence Limitation	The provided screenshot confirms the visible hierarchy and security accounts. The user report is used to document the full OU list when an OU is outside or partially outside the captured view.
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Architecture



Enterprise-style root hierarchy separates security, infrastructure, workloads, exceptions, lifecycle, sandbox, and incident-response functions.

Figure 1 - Enterprise-style AWS Organizations hierarchy built in this project.

Implementation Summary

1. Signed in through the IAM Identity Center SSO access path established in previous projects.
2. Opened AWS Organizations from the management account context.
3. Selected the organization root as the target for new top-level OUs.
4. Created Infrastructure for shared infrastructure and operations services.
5. Created Workloads for future application accounts.
6. Created Exceptions for temporary accounts that need to operate outside normal guardrails.
7. Created Sandbox for accounts where experimentation can be permitted with looser rules.
8. Created Onboarding for accounts being adopted or moved into the organization.
9. Created Nursery for newly created accounts before they move into their final OU.

- 10. Created Suspended for accounts being prepared for deprovisioning or lock-down.
- 11. Created IncidentResponse for accounts requiring special controls during active incident work.
- 12. Created new AWS accounts named IAM and SecurityOperations.
- 13. Placed IAM, SecurityAudit, and SecurityOperations under the Security OU.

GOVERNANCE

This project is a governance foundation. The OU structure prepares the organization for future SCP layering, permission boundaries, delegated administration, account lifecycle workflows, and workload separation.

OU Purpose Map

Security	OU	Security governance, audit, identity, and operations accounts.
Infrastructure	OU	Shared networking, operations, and infrastructure services.
Workloads	OU	Future application and environment accounts.
Exceptions	OU	Temporary deviations from standard guardrails.
Sandbox	OU	Developer or lab experimentation with controlled freedom.
Onboarding	OU	Accounts being adopted from another organization or transition path.
Nursery	OU	Newly created accounts before final placement and hardening.
Suspended	OU	Accounts being decommissioned or tested for shutdown impact.
IncidentResponse	OU	Accounts under active incident-response controls.
IAM	Account	Future management location for IAM Identity Center and identity administration.
SecurityOperations	Account	Future location for security tooling that may make changes.
SecurityAudit	Account	Existing audit and assessment account.

Evidence

Figure 2 - Enterprise OU Structure and Security Accounts



The redacted AWS Organizations hierarchy evidence shows the root with multiple top-level OUs, including Exceptions, IncidentResponse, Infrastructure, Nursery, Onboarding, Sandbox, Security, and Suspended. The Security OU is expanded and shows IAM, SecurityAudit, and SecurityOperations accounts. Sensitive identifiers were redacted before publication.

Security Analysis

The expanded OU hierarchy creates governance boundaries that can receive different Service Control Policies based on account purpose and risk.

Placing IAM and SecurityOperations under Security separates high-impact identity and security tooling from general workloads.

The SecurityAudit and SecurityOperations split supports separation between read-oriented audit or assessment capabilities and tooling that may perform write or response actions.

Lifecycle OUs such as Nursery, Onboarding, Suspended, and Exceptions help manage accounts before, during, and after normal production use rather than forcing every account into the same policy path.

An IncidentResponse OU gives the organization a place to apply unusual or urgent guardrails without redesigning normal workload governance during an active incident.

Lessons Learned

AWS accounts are security and operational boundaries. Organizing them intentionally helps reduce blast radius and improves accountability.

OU design should reflect function, ownership, lifecycle state, and expected guardrails rather than just application names.

SCP strategy often drives OU design because policies can be attached at the OU level and inherited by accounts underneath.

Enterprise structures evolve. This project creates the top-level structure, while workload sub-OUs, production/non-production separation, and refined SCPs can be added later.

Production Improvements

- Add Prod and NonProd sub-OUs under Workloads when workload accounts are introduced.
- Create and test separate SCP baselines for Workloads, Sandbox, Suspended, Exceptions, and IncidentResponse.
- Define account vending standards for accounts that enter Nursery.
- Assign clear ownership, cost center, purpose, and lifecycle metadata for every account.
- Plan delegated administration for IAM, SecurityOperations, and SecurityAudit functions.
- Create a runbook for moving accounts between Onboarding, Nursery, Workloads, Suspended, and IncidentResponse.

Skills Demonstrated

AWS Organizations	Blast-radius reduction	OU hierarchy design
Account isolation	SCP planning	Security account separation
Account lifecycle design	Enterprise structure	Multi-account administration

Resume Bullet

RESUME

Expanded an AWS Organizations environment into an enterprise-style hierarchy by creating multiple top-level OUs for infrastructure, workloads, exceptions, sandbox, onboarding, nursery, suspended, and incident-response workflows, and by adding IAM and SecurityOperations accounts under the Security OU.

STAR Interview Story

Situation: The AWS organization had core security services configured but still needed a scalable account hierarchy for enterprise-style governance.

Task: Create a root-level OU structure and add dedicated accounts for identity and security operations.

Action: I created top-level OUs for infrastructure, workloads, exceptions, sandbox, onboarding, nursery, suspended, and incident response, then created IAM and SecurityOperations accounts and placed them under the Security OU with SecurityAudit.

Result: The environment now has a more mature multi-account structure that supports future SCP layering, account lifecycle workflows, and separation of security responsibilities.

Version History

v1.0.0	July 13, 2026	Initial documented implementation. Created enterprise-style top-level OUs, created IAM and SecurityOperations accounts, placed security accounts under the Security OU, documented evidence limitations, and embedded provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - AWS LEGO: Organizing the Org.
- User-provided redacted AWS Organizations hierarchy screenshot.
- Project 08 - AWS Organizations Security OU and SecurityAudit Account.
- Project 09 - Organization-Wide Service Control Policies.
- Project 14 - IAM Identity Center AdministratorAccess Assignment.